IE 7275 Data Mining in Engineering
Homework 04
Deadline: 12/11/2022
Total point is 110 points

- **Submit all necessary files along with your solution sheets on Canvas.**
- **You can use Python or equivalent programming languages.**

Problem 1 (CART) [15 points]

Given the training dataset in **Table 1** and the testing dataset in **Table 2**, answer the following questions:

(a) Build the complete decision tree using binary splits and information gain as the evaluation measure.

(b) Compute the accuracy of the classifier on the test data. Also show the per class accuracy and coverage.

<u>Note that you are required to elaborate all steps towards a final DT.</u>

Table 1 and Table 2

X	*Y*	*Z*	Class
15	1	*A*	1
20	3	*B*	2
25	2	*A*	1
30	4	*A*	1
35	2	*B*	2
25	4	*A*	1
15	2	*B*	2
20	3	*B*	2

(a) Training

X	*Y*	*Z*	Class
10	2	*A*	2
20	1	*B*	1
30	3	*A*	2
40	2	*B*	2
15	1	*B*	1

(b) Testing

Problem 2 (Clustering Analysis) [10 points]

Given the data points in **Table 3**, and their probability of belonging to two clusters. Assume that these points were produced by a mixture of two univariate normal distributions. Answer the following questions:

(a) Find the maximum likelihood estimate of the means $\mu 1$ and $\mu 2$.

(b) Assume that $\mu 1 = 2$, $\mu 2 = 7$, and $\sigma 1 = \sigma 2 = 1$. Find the probability that the point $x = 5$ belongs to cluster C1 and to cluster C2. You may assume that the prior probability of each cluster is equal (i.e., P (C1) = P (C2) = 0.5), and the prior probability P (x = 5) = 0.029.

<u>Note that you are required to elaborate all steps.</u>

Table 3

x	$P(C_1\|x)$	$P(C_2\|x)$
2	0.9	0.1
3	0.8	0.1
7	0.3	0.7
9	0.1	0.9
2	0.9	0.1
1	0.8	0.2

Problem 3 (Clustering Analysis) [15 points]

The dataset **Cereals.csv** includes nutritional information, store display, and consumer ratings for 77 breakfast cereals. Data preprocessing. Remove all cereals with missing values.

1. Apply hierarchical clustering to the data using Euclidean distance to the normalized measurements. Compare the dendrograms from single linkage and complete linkage, and look at cluster centroids. Comment on the structure of the clusters and on their stability.
   (Hint: To obtain cluster centroids for hierarchical clustering, compute the average values of each cluster members, using groupby() with the cluster centers followed by mean: dataframe.groupby(clusterlabel).mean().)
2. Which method leads to the most insightful or meaningful clusters?
3. Choose one of the methods. How many clusters would you use? What distance is used for this cutoff? (Look at the dendrogram.)
4. The elementary public schools would like to choose a set of cereals to include in their daily cafeterias. Every day a different cereal is offered, but all cereals should support a healthy diet. For this goal, you are requested to find a cluster of "healthy cereals." Should the data be normalized? If not, how should they be used in the cluster analysis?

Problem 4 (Clustering Analysis) [10 points]

Given a distance matrix among five members {a, b, c, d, e} in a feature space in **Table 4**, develop your hierarchical clustering approach with <u>complete linkage</u> to find the relationships among them (i.e., show a hierarchical structure).

Table 4

	a	b	c	d	e
a	0				
b	9	0			
c	3	7	0		
d	6	5	9	0	
e	11	10	2	8	0

Problem 5 (Association Rules) [15 points]

Consider a dataset on cosmetic purchases (**Cosmetics.csv**) at a large chain drugstore. The store wants to analyze associations among purchases of these items for purposes of point-of-sale display, guidance to sales personnel in promoting cross-sales, and guidance for piloting an eventual time-of-purchase electronic recommender system to boost cross-sales. Consider first only the data shown in Table 14.15, given in binary matrix form.

(a) Consider a subset of the data shown in **Table 5** and the partial results of association rules analysis shown in **Table 6**. Explain the outputs of "confidence", "support", and "lift" and how they are calculated.

(b) Now, use the complete dataset on the cosmetics purchases (in the file **Cosmetics.csv**). Using Python, apply association rules to these data (for apriori use min_support=0.1 and use_colnames=True, for association_rules use default parameters).

1) Report association rules as shown in Table 6. Select top 20 rules in terms of high confidence and lift.
2) Reviewing the resulting rules, comment on their redundancy and how you would assess their utility.

Table 5

Trans. #	Bag	Blush	Nail polish	Brushes	Concealer	Eyebrow pencils	Bronzer
1	0	1	1	1	1	0	1
2	0	0	1	0	1	0	1
3	0	1	0	0	1	1	1
4	0	0	1	1	1	0	1
5	0	1	0	0	1	0	1
6	0	0	0	0	1	0	0
7	0	1	1	1	1	0	1
8	0	0	1	1	0	0	1
9	0	0	0	0	1	0	0
10	1	1	1	1	0	0	0
11	0	0	1	0	0	0	1
12	0	0	1	1	1	0	1

Table 6

```
   lhs                  rhs                 support   confidence        lift
1 {Blush,
   Concealer,
   Mascara,
   Eye.shadow,
   Lipstick}    => {Eyebrow.Pencils}   0.013  0.3023255814  7.198228128
2 {Trans.,
   Blush,
   Concealer,
   Mascara,
   Eye.shadow,
   Lipstick}    => {Eyebrow.Pencils}   0.013  0.3023255814  7.198228128
3 {Blush,
   Concealer,
   Mascara,
   Lipstick}    => {Eyebrow.Pencils}   0.013  0.2888888889  6.878306878
```

Problem 6 (Association Rules) [10 points]

You are given a transaction database for a retail shop in **Table 7**.

(a) Compute the support for item set {Milk, Rice, Soft Drink} and compute confidence, lift ratio for rule {Milk, Soft Drink} ➔ Rice

(b) Compute support, confidence, and lift ration for the rules Rice ➔ Chicken and Apple ➔ Pear and comment on their strengths.

Table 7

TID	Apple	Pear	Milk	Rice	Chicken	Soft Drink
1	1	0	0	1	0	1
2	1	0	0	0	0	1
3	0	0	1	1	0	1
4	0	1	1	0	0	0
5	1	0	1	1	1	0
6	0	1	0	0	0	1
7	0	0	1	0	0	1
8	1	0	0	1	1	1
9	1	1	0	0	0	0
10	0	0	1	1	1	1

Problem 7 (Collaborative Filtering) [20 points]

The Institute for Statistics Education at Statistics.com asks students to rate a variety of aspects of a course as soon as the student completes it. The Institute is contemplating instituting a recommendation system that would provide students with recommendations for additional courses as soon as they submit their rating for a completed course. Consider the excerpt from student ratings of online statistics courses shown in **Table 8**, and the problem of what to recommend to student E.N.

(a) First consider a user-based collaborative filter. This requires computing correlations between all student pairs. For which students is it possible to compute correlations with E.N.? Compute them.

(b) Based on the single nearest student to E.N., which single course should we recommend to E.N.? Explain why.

(c) Use scikit-learn function sklearn.metrics.pairwise.cosine_similarity() to compute the cosine similarity between users.

(d) Based on the cosine similarities of the nearest students to E.N., which course should be recommended to E.N.?

(e) Apply item-based collaborative filtering to this dataset (using Python) and based on the results, recommend a course to E.N.

Table 8

	SQL	Spatial	PA 1	DM in R	Python	Forecast	R Prog	Hadoop	Regression
L N	4				3	2	4		2
M H	3	4			4				
J H	2	2							
E N	**4**			**4**			**4**		**3**
D U	4	4							
F L		4							
G L		4							
A H		3							
S A			4						
R W			2					4	
B A			4						
M G			4			4			
A F			4						
K G			3						
D S	4			2			4		

Problem 8 (Collaborative Filtering) [15 points]

See the dataset below.

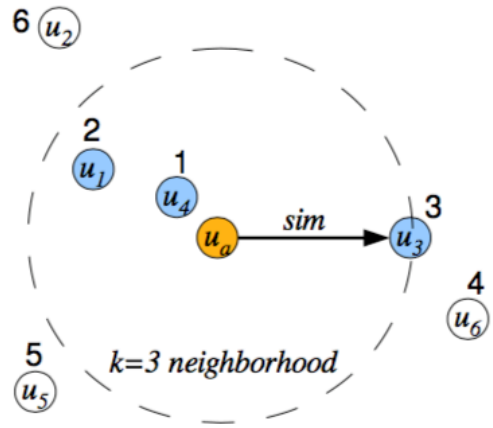
(a) Conduct a <u>user-based</u> collaborative filtering approach to predict the ratings (the last row in yellow) of items 1, 2, 5, and 7 for user u_a. The rating may be estimated differently based on your choice of similarity measure. Specify your settings such as k and similarity measure.

(b) Conduct an <u>item-based</u> collaborative filtering approach to predict the ratings (the last row in yellow) of items 1, 2, 5, and 7 for user u_a. Specify your settings such as k and similarity measure.

Highlighted rows in blue are just for showing what are considered as nearest neighbors, and do not mean the solutions.

	i_1	i_2	i_3	i_4	i_5	i_6	i_7	i_8
u_1	?	4.0	4.0	2.0	1.0	2.0	?	?
u_2	3.0	?	?	?	5.0	1.0	?	?
u_3	3.0	?	?	3.0	2.0	2.0	?	3.0
u_4	4.0	?	?	2.0	1.0	1.0	2.0	4.0
u_5	1.0	1.0	?	?	?	?	?	1.0
u_6	?	1.0	?	?	1.0	1.0	?	1.0
u_a	?	?	4.0	3.0	?	1.0	?	5.0
$\hat{r}_a$	3.5	4.0			1.3		2.0	

(a)



(b)